

MAT 425 HW # 8

In function spaces, the inner product of two functions $f(x)$ and $g(x)$ over the interval $[-1, 1]$ is defined as:

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx.$$

1. (4 points) check if the functions $f_1(x) = 1$, $f_2(x) = \sin(\pi x)$, $f_3(x) = \cos(\pi x)$, are orthogonal.

The norm of a function measures its size and is given by:

$$\|f\| = \sqrt{\langle f, f \rangle}.$$

This is analogous to the length of a vector in Euclidean space.

2. (3 points) Compute the norms $\|f_1\|$, $\|f_2\|$, and $\|f_3\|$.
3. (2 points) What do we mean by an orthonormal set of functions?
4. (1 point) Create an orthonormal set using the given functions $f_1, \dots, f_3$.